

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Mock Interview

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	D.Sravanthi	Reg. No.:	192425315
Section / Batch:	2024-AI&ML	Date:	17.08.2026

Code of Conduct

I __D.Sravanthi(192425315)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __D.Sravanthi____

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical &	Real-world scenario involving discourse analysis, conversational	Q10

Annexure A – Mock Interview

CHATGPT PROMPT: copy and paste in your chatgpt

MOCK INTERVIEW ASSESSMENT – NLP (Language Generation and Discourse Analysis)

Act as an expert Technical Interview Panel for Computer Science and Engineering students.

Interview Domain:

Language Generation and Discourse Analysis

Topics:

- Discourse
- Reference Resolution
- Text Coherence
- Discourse Structure
- Coherence
- Dialog and Conversational Agents
- Dialog Acts
- Interpretation
- Language Generation
- Architecture
- Surface Realization
- Discourse Planning
- Machine Translation
- Transfer Metaphor
- Interlingua
- Statistical Approaches

Interview Instructions:

Step 1:

Ask the student to enter:

Student Name:

Register Number:

Step 2:

Conduct a technical interview consisting of exactly 10 questions.

Rules:

- Ask only one question at a time.
- Wait for the student's response.
- Ask conceptual, analytical, application-based, and problem-solving questions.
- Include at least:
 - 2 questions on Discourse and Coherence
 - 2 questions on Conversational Agents and Dialog Acts
 - 2 questions on Language Generation
 - 2 questions on Machine Translation

- 2 analytical or scenario-based questions

Step 3:

Store every question and answer.

Step 4:

After Question 10, evaluate the student using the following rubric.

RUBRIC

Technical Knowledge (25)

Level 4:

Demonstrates strong and accurate subject knowledge with in-depth understanding

Level 3:

Shows good knowledge with minor gaps

Level 2:

Basic knowledge with noticeable errors

Level 1:

Lacks fundamental technical knowledge

Problem Solving (25)

Level 4:

Solves problems logically and applies concepts effectively in real scenarios

Level 3:

Applies concepts with moderate accuracy

Level 2:

Limited problem-solving ability; requires guidance

Level 1:

Unable to solve or apply concepts

Analytical Thinking (20)

Level 4:

Analyzes questions critically and provides well-structured logical answers

Level 3:

Provides reasonable analysis with some structure

Level 2:

Superficial or partially structured responses

Level 1:

No analytical thinking demonstrated

Communication (15)

Level 4:

Communicates clearly, confidently and professionally using appropriate terminology

Level 3:

Communicates well with minor hesitation

Level 2:

Communication lacks clarity or confidence

Level 1:

Unable to communicate effectively

Confidence (15)

Level 4:

Displays high confidence, positive attitude and professional behavior throughout

Level 3:

Shows good confidence with minor issues

Level 2:

Low confidence and inconsistent behavior

Level 1:

Lacks confidence and professionalism

MARK CALCULATION

Level 4 = 100% of weightage

Level 3 = 75% of weightage

Level 2 = 50% of weightage

Level 1 = 25% of weightage

Technical Knowledge = 25 marks

Problem Solving = 25 marks

Analytical Thinking = 20 marks

Communication = 15 marks

Confidence = 15 marks

Calculate:

Final Score = Sum of all rubric scores

Step 5:

Generate the final report in the exact format below.

MOCK INTERVIEW ASSESSMENT REPORT

Student Name:

Register Number:

INTERVIEW QUESTIONS AND RESPONSES

Q1:

Question Asked:

Student Response:

Q2:
Question Asked:
Student Response:

Q3:
Question Asked:
Student Response:

Q4:
Question Asked:
Student Response:

Q5:
Question Asked:
Student Response:

Q6:
Question Asked:
Student Response:

Q7:
Question Asked:
Student Response:

Q8:
Question Asked:
Student Response:

Q9:
Question Asked:
Student Response:

Q10:
Question Asked:
Student Response:

RUBRIC EVALUATION

Criterion	Weightage Assigned	Level	Marks Awarded
Technical Knowledge	25	Level X	XX
Problem Solving	25	Level X	XX
Analytical Thinking	20	Level X	XX
Communication	15	Level X	XX
Confidence	15	Level X	XX

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	SubQ Total	Student Avg Final Mark
Level X	Level X	Level X	Level X	Level X	XX.X	XX.X

OVERALL FEEDBACK

Strengths:

- List strengths

Areas for Improvement:

- List improvements

Interview Readiness:

- Excellent / Good / Average / Needs Improvement

End the report after the Interview Readiness section.

After evaluating each answer, do NOT display the marks immediately.

Store the evaluation internally.

After Question 10 is completed:

1. Generate a Question-wise Evaluation Table for all 10 questions.

2. For each question assign:

- Technical Knowledge Level
- Problem Solving Level
- Analytical Thinking Level
- Communication Level
- Confidence Level
- SubQ Total

3. Calculate SubQ Total using:

Level 4 = 100% weightage

Level 3 = 75% weightage

Level 2 = 50% weightage

Level 1 = 25% weightage

4. Calculate Student Avg Final Mark as:

Average of all 10 SubQ Totals

5. Display:

- Complete interview questions
- Student answers
- Question-wise evaluation table
- Student Avg Final Mark
- Overall strengths
- Areas for improvement
- Interview readiness

Student Name: Sravanthi Duggirala

Register Number: 192425315

Q1

Question Asked: What is discourse in NLP, and why is discourse coherence important?

Student Response:

Discourse refers to a connected sequence of sentences or utterances that together convey a meaningful message. In NLP, discourse helps us understand how sentences are related to each other. Discourse coherence is important because it makes the text logically connected and understandable. For example, in “Ravi went to the library. He borrowed a book,” the second sentence is connected to the first through the reference “He.” Without coherence, the text may contain grammatically correct sentences but the overall meaning may be unclear.

Q2

Question Asked: Consider the sentence: “Priya gave Anjali her book because she needed it.” Identify the possible ambiguities and explain how reference resolution can solve them.

Student Response:

The pronouns “her,” “she,” and “it” can create ambiguity. “Her” could refer to Priya or Anjali, while “she” could also refer to either person. “It” most likely refers to the book. Reference resolution uses information such as gender, number, grammatical role, recency, and semantic context to identify the correct antecedent. A complete system may also need wider discourse context to determine the intended meaning.

Q3

Question Asked: What is a conversational agent? Explain the role of dialog acts in conversational systems.

Student Response:

A conversational agent is an NLP-based system that communicates with users through natural language, either by text or speech. Examples include chatbots, virtual assistants, and customer-support systems. Dialog acts represent the function or intention of an utterance, such as greeting, requesting, questioning, informing, confirming, or thanking. Identifying dialog acts helps the system understand what the user wants and select an appropriate response.

Q4

Question Asked: A user says, “Can you book me a flight to Delhi tomorrow?” What dialog act is represented, and how should a conversational agent respond?

Student Response:

The main dialog act is a request because the user is asking the system to perform an action. The conversational agent should identify the intent as flight booking and extract important information such as destination and date. If required information such as departure city, passenger count, or preferred time is missing, the system should ask a clarification question. After obtaining all necessary information, it can search for suitable flights and confirm the booking details.

Q5

Question Asked: Explain the architecture of a Natural Language Generation system. What are discourse planning and surface realization?

Student Response:

A Natural Language Generation system converts structured information into natural-language text. A common architecture includes content determination, discourse planning, sentence planning, and surface realization. Discourse planning decides what information should be communicated and in what order. Sentence planning organizes the selected information into sentences and chooses appropriate expressions. Surface realization converts the planned representation into grammatically correct natural-language sentences.

Q6

Question Asked: Suppose an AI system receives weather data: “Temperature: 32°C, Humidity: 80%, Rain probability: 70%.” Generate a suitable natural-language response and explain the generation process.

Student Response:

A suitable response would be: “Today will be hot and humid, with a high probability of rain. The temperature is around 32°C and the chance of rain is 70%.”

The system first determines the important information from the input data. During discourse planning, it decides to mention the temperature, humidity, and rain probability. During sentence planning, related information is combined into meaningful sentences. Finally, surface realization generates grammatically correct natural language.

Q7

Question Asked: What is Machine Translation? Explain the Transfer approach and Interlingua approach.

Student Response:

Machine Translation is the automatic conversion of text from one language into another. In the Transfer approach, the source-language representation is analyzed and transformed into an intermediate representation that is specific to the source and target languages. Then the target-language sentence is generated. In the Interlingua approach, the source sentence is converted into a language-independent semantic representation called an interlingua. The target language is then generated from this representation. Interlingua can be useful for multilingual systems because the same representation can support multiple target languages.

Q8

Question Asked: A company wants to translate documents between English, Tamil, Hindi, and Telugu. Which approach would be more scalable: Transfer or Interlingua? Explain your reasoning.

Student Response:

Interlingua can be more scalable for a large number of languages because each language is connected to a common language-independent representation. With a transfer-based system, separate language-pair

transfer components may be required, which becomes complex as the number of languages increases. Interlingua reduces the number of direct language-pair relationships. However, developing an accurate language-independent semantic representation is difficult and requires significant linguistic knowledge.

Q9

Question Asked: A chatbot answers every user question correctly at the sentence level, but users complain that the conversation feels disconnected and unnatural. Analyze the possible problem and suggest solutions.

Student Response:

The problem may be related to discourse management and coherence rather than sentence-level correctness. The chatbot may not be maintaining conversation history, user intent, entities, or topic continuity. It may also fail to use appropriate dialog acts and reference previous statements. To improve it, the system should maintain dialog state, track entities and references, identify dialog acts, use discourse context, and generate responses that connect naturally with previous turns.

Q10

Question Asked: A machine translation system produces grammatically correct sentences but often changes the intended meaning of the original text. What could cause this problem, and how would you improve the system?

Student Response:

The problem may occur because the system focuses mainly on grammatical or word-level translation without properly capturing context, discourse meaning, ambiguity, and long-distance dependencies. Polysemous words may also be translated incorrectly when context is ignored. I would improve the system by using contextual representations, larger and better-quality parallel datasets, discourse-aware processing, reference resolution, and context-sensitive translation models. Evaluation should consider semantic adequacy as well as grammatical fluency.

Question-Wise Evaluation

Question	Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence	SubQ Total
Q1	Level 4	Level 4	Level 4	Level 4	Level 4	100
Q2	Level 4	Level 4	Level 4	Level 4	Level 4	100

Question	Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence	SubQ Total
Q3	Level 4	Level 4	Level 4	Level 4	Level 4	100
Q4	Level 4	Level 4	Level 4	Level 4	Level 4	100
Q5	Level 4	Level 4	Level 4	Level 4	Level 4	100
Q6	Level 4	Level 4	Level 4	Level 4	Level 4	100
Q7	Level 4	Level 4	Level 4	Level 4	Level 4	100
Q8	Level 4	Level 4	Level 4	Level 4	Level 4	100
Q9	Level 4	Level 4	Level 4	Level 4	Level 4	100
Q10	Level 4	Level 4	Level 4	Level 4	Level 4	100

Rubric Evaluation

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level 4	25
Problem Solving	25	Level 4	25
Analytical Thinking	20	Level 4	20
Communication	15	Level 4	15
Confidence	15	Level 4	15
Total	100	Level 4	100

Summary Table

Criterion	Level	Marks
Technical Knowledge (25)	Level 4	25
Problem Solving (25)	Level 4	25
Analytical Thinking (20)	Level 4	20
Communication (15)	Level 4	15
Confidence (15)	Level 4	15
SubQ Total	Level 4	100
Student Avg Final Mark	Level 4	100.0

Overall Feedback

Strengths:

- Demonstrates strong understanding of discourse and coherence.
- Correctly explains reference resolution and ambiguity.
- Understands conversational agents and dialog acts.
- Clearly explains NLG architecture, discourse planning, and surface realization.
- Provides appropriate application-based examples.
- Demonstrates good understanding of Transfer and Interlingua approaches.
- Provides logical solutions to analytical and scenario-based questions.
- Uses appropriate NLP terminology.

Areas for Improvement:

- Could provide more detailed examples of statistical machine translation.
- Could explain specific algorithms and models used in modern NLP systems.
- Could discuss limitations of Interlingua and Transfer approaches in greater depth.

Interview Readiness:

Excellent

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____